

Phase 3.2 Contact Audit Report

Date: 2026-07-16

1. Probability Distribution Analysis

Recomputed directly from `contact_predictions.pt` (N = 230,753 atomic pairs):

- **Minimum Probability:** 0.003052
- **Maximum Probability:** 0.078176
- **Mean Probability:** 0.007511
- **Median Probability:** 0.005893
- **Standard Deviation:** 0.004481

Percentiles

- **10th percentile:** 0.004074
- **25th percentile:** 0.004622
- **50th (Median):** 0.005893
- **75th percentile:** 0.008777
- **90th percentile:** 0.013194
- **95th percentile:** 0.016143
- **99th percentile:** 0.025018

2. Resolution of the Threshold Mismatch

- **Under the Default Threshold ($t = 0.5$):**

- Positive Predictions: **0** - Negative Predictions: **230,753** - Precision: **0.0000** - Recall: **0.0000** - F1: **0.0000**

- **Under the Prior Probability Threshold ($t = 0.00587$):**

- Positive predictions matches class prior, yielding non-zero outputs but low precision (**0.0122**) due to massive background class imbalance.

Discrepancy Cause

The original `Contact Analysis Report` contained **stale cached placeholder values** (Precision = 0.814, Recall = 0.763, F1 = 0.788) from Phase 3.1C's IRM template. These have been successfully regenerated and corrected.

3. Auxiliary Head Learnability Status

The auxiliary head has **failed to learn** (PR-AUC = 0.0122, barely above the random baseline of 0.00587). This is due to severe class imbalance (0.58% positive contacts) without positive weight scaling or prior bias initialization, causing the network output to collapse to prior levels.